1 Explanation certificate: churn100-s4-c3

This section is the explanation certificate of the churn100-s4-c3 network. It is rendered from the canonical JSON record `churn100-s4-c3.json` (schema 1.4.0, SHA-256 `99f3bce0c6f3eff7...`), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 1 summarises the network and its coverage.

Dataset	churn_onehot
Experiment	churn100-s4, seed 4
Genome	`27f540ac-6acb-4d58-9972-cd71fae31cf0` (NEAT key 24445, generation 99)
Explanation	`ffdb207f-1e0f-4d2e-a330-f47b1ca75c4a` (89 operations)
Train AUC / accuracy	0.8884 / 0.8795 (n = 4000)
Test AUC / accuracy	0.8671 / 0.875 (n = 1000)
Output activation	sigmoid
Phenotype	93 nodes (71 inputs, 21 hidden, 1 output), 76 enabled edges
Explained model state	103 nodes, 78 enabled edges
Refactoring operations	add_identity_node 2, consolidate_node 1, split_node 8
Function preservation	max $\lvert\Delta\,\text{output}\rvert = 4.44 \times 10^{-16}$ on the training split
Annotations	7 (5 leaf, 2 composition)
Structural coverage	1 (102 of 102 candidate nodes; 30 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`99f3bce0c6f3eff7...`

Table 1: Explanation certificate summary: churn100-s4-c3.

Drafter's summary. logit P(churn) = -1.35 + a sparse linear backbone over 28 inputs (voicemail plan -1.05 per SD, international plan +0.80, voicemail messages +0.68, evening charge +0.40, state offsets) + a customer-service-calls threshold (an early hinge from 3 calls, off in state:29, plus a late chain from 5 calls gated by state:4/state:5: nothing for 0-2 calls, then about +2.3 logit per call from 4) + three day-usage hinges (nothing below about 200 day minutes, then rising in stages to +4.7 logit above 300 minutes; the first hinge starts about 54 minutes later with a voicemail plan) + exact single-state offsets, all additive in the logit. Day minutes and service calls have no linear term. Train AUC 0.888, accuracy 0.880. Compared with the run's cluster-1 lead (churn100-s4, key 24573), the same two thresholds (service calls, high day usage) carry the nonlinearity, but here day usage is gated by the voicemail plan rather than combined with evening minutes, there is no evening term beyond the linear charge, and many more single-state units.

What resisted explanation. Nothing structural: no hidden node needed splitting (8 input splits, 1 consolidation of two customer-service-calls copies). 11 of the 21 hidden nodes are single-state offsets (exactly affine in one-hot indicators) and 5533 is never active on train (contributes exactly 0), so the nonlinearity sits in 8 nodes: the service-call hinges and the day-usage hinges. Near-cancelling state pairs: state:39 direct -2.85 vs unit +2.30 (net -0.55, empirical -0.93), state:46 -2.22 vs +0.77, state:33 +1.05 vs -0.68. The model is badly calibrated in level: mean prediction 0.082 against a base rate of 0.1415 (0-2 calls: 0.05-0.06 vs 0.11; 4 calls 0.21 vs 0.44), and over-confident from 6 calls (0.89-0.98 vs 0.55-0.68, 42 customers). state:18 gets +0.36 logit against an empirical -0.48 (65 rows); state:4 switches off the late service-call chain although its 41 customers

churn more (empirical +0.80). closed_form_max_abs_err 0.028 (state_offsets) is 3-significant-figure rounding; numeric error 1.8e-15.

1.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, no voice-mail messages, 1 call to customer service and median usage) the model predicts a chance of leaving of about 1%; it adds up four parts: fixed-weight inputs, service calls, heavy daytime use and state units. [C054] This part adds up most inputs with fixed weights: an international plan raises a typical customer's chance of leaving from about 1% to about 18%, and a voice-mail plan lowers it to below 1%. [C055, C063] More voice-mail messages raise it steadily, from about 1% at 0 to about 6% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C056] A higher evening charge also raises it steadily, from about 1% at 11.76 US dollars to about 2% at 22.57 US dollars, flattening out near the bottom because the predicted chance cannot go much lower. [C059] Through it, the state moves the chance from about 1% in West Virginia to about 5% in Washington and about 4% in Nevada, and to below 1% in Rhode Island, Hawaii and Vermont (72 to 82 training customers per state). [C057, C058, C061, C062, C064] This part adds up two parts: an early rise and a late rise in calls to customer service. [C090] From about 3 calls upwards it adds to the chance slowly at first, then faster: for a typical such customer, 4 calls take it from about 1% to about 10%, flattening out near the bottom because the predicted chance cannot go much lower. [C089] Up to 3 calls the chance rises slowly at first, then faster, from about 1% to about 2% for a typical customer, flattening out near the bottom because the predicted chance cannot go much lower. [C084] That early rise happens only outside Nebraska, California and Colorado: in Nebraska the chance stays at about 1% from 0 to 3 calls, against a rise from about 1% to about 2% elsewhere, slowly at first, then faster, flattening out near the bottom because the predicted chance cannot go much lower. [C088] Daytime minutes add to the chance only once they rise above about 209 minutes, in an uneven way: for a typical such customer, from about 1% at 0 minutes to about 16% at 271 minutes, flattening out near the bottom because the predicted chance cannot go much lower. [C100] The rise is much larger without a voice-mail plan: from 112 to 249 daytime minutes the chance goes from about 1% to about 6% without one, but stays below 1% with one, in an uneven way, flattening out near the bottom because the predicted chance cannot go much lower. [C099] This part acts on 1074 of 4000 training customers, 290 of whom left; for the typical customer it acts on, the chance is about 1% without it and about 4% with it. [C102] This part reads only the state: it raises a typical customer's chance from about 1% to about 12% in Rhode Island, to about 3% in Vermont and to about 2% in Tennessee. [C104, C105, C106] It lowers the chance to below 1% in Virginia and in Nevada. [C107, C108]

The whole model. The model adds up four parts into one score: fixed-weight inputs, service calls, heavy daytime use and state units. [C054] *Evidence.* It is made up of these four parts (each explained below); the score is their sum, which was checked on all 4000 training customers. [C054] Changing one input at a time across the whole model, an international plan takes a typical customer's chance of leaving from about 1% to about 18%, and a voice-mail plan lowers it to below 1%. [C001, C008] More voice-mail messages raise it steadily, from about 1% at 0 to about 6% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C002] More daytime minutes raise it in an uneven way, from about 1% at 112 minutes to about 6% at 249 minutes, flattening out near the bottom because the predicted chance cannot go much lower. [C003] The model ignores evening calls, evening minutes, international calls, the international charge, international minutes,

night-time calls, the night-time charge and night-time minutes. [C046, C047, C048, C049, C050, C051, C052, C053]

Fixed-weight inputs. This part adds up most inputs with fixed weights: an international plan, voice-mail messages and the evening charge raise the chance of leaving, steadily for the numbers, and a voice-mail plan lowers it. [C055, C056, C059, C063] *Evidence.* Through it, an international plan takes a typical customer from about 1% to about 18%, and a voice-mail plan lowers the chance to below 1%. [C055, C063] More voice-mail messages raise it steadily, from about 1% at 0 to about 6% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C056] A higher evening charge also raises it steadily, from about 1% at 11.76 US dollars to about 2% at 22.57 US dollars, flattening out near the bottom because the predicted chance cannot go much lower. [C059] Through it, the state moves the chance from about 1% in West Virginia to about 5% in Washington and about 4% in Nevada, and to below 1% in Rhode Island, Hawaii and Vermont (72 to 82 training customers per state). [C057, C058, C061, C062, C064] Its formula reproduces the part exactly on all 4000 training customers. [C083]

Service calls. This part raises the chance of leaving with calls to customer service, slowly at first, then faster from about 3 calls upwards; it adds up two parts. [C090, C089] *Evidence.* It is made of two parts that are added: an early rise and a late rise in calls to customer service. [C090] From about 3 calls upwards it adds to the chance slowly at first, then faster: for a typical such customer, 4 calls take it from about 1% to about 10%, flattening out near the bottom because the predicted chance cannot go much lower. [C089] Up to 3 calls the chance rises slowly at first, then faster, from about 1% to about 2% for a typical customer, flattening out near the bottom because the predicted chance cannot go much lower. [C084] That early rise happens only outside Nebraska, California and Colorado: in Nebraska the chance stays at about 1% from 0 to 3 calls, against a rise from about 1% to about 2% elsewhere, slowly at first, then faster, flattening out near the bottom because the predicted chance cannot go much lower. [C088]

Early rise in calls. This part raises the chance of leaving from about 3 customer-service calls upwards, in an uneven way. [C094] *Evidence.* It acts on 830 of 4000 training customers, 213 of whom left; for the typical customer it acts on, the chance is about 1% without it and about 2% with it. [C095] Through it, calls add to the chance only from about 3 calls upwards, in an uneven way: for a typical such customer, 4 calls take it from about 1% to about 10%, flattening out near the bottom because the predicted chance cannot go much lower. [C094] Up to 3 calls it raises the chance slowly at first, then faster, but only outside Nebraska: there from about 1% to about 2%, while in Nebraska it stays at about 1%, flattening out near the bottom because the predicted chance cannot go much lower. [C093] Its formula reproduces the part exactly on all 4000 training customers. [C096]

Late rise in calls. This part raises the chance of leaving for a small group of customers, more in Colorado and less in California. [C122, C117, C118] *Evidence.* It acts on 134 of 4000 training customers, 74 of whom left; for the typical customer it acts on, the chance is about 39% without it and about 75% with it. [C122] For the typical customer it acts on, the chance through it is about 75% in Colorado and about 39% in California, against about 52% in other states. [C117, C118] Its formula reproduces the part exactly on all 4000 training customers. [C123]

Heavy daytime use. This part raises the chance of leaving for heavy daytime users: daytime minutes add to it only above about 209 minutes, in an uneven way, and much less with a voice-mail plan. [C100, C099] *Evidence.* This part acts on 1074 of 4000 training customers, 290 of whom left; for the typical customer it acts on, the chance is about 1% without it and about 4% with it. [C102] Daytime minutes add to the chance only once they rise above about 209 minutes, in an uneven way: for a typical such customer, from about 1% at 0 minutes to about 16% at 271 minutes, flattening out near the bottom because the predicted chance cannot go much lower. [C100] The rise is much

larger without a voice-mail plan: from 112 to 249 daytime minutes the chance goes from about 1% to about 6% without one, but stays below 1% with one, in an uneven way, flattening out near the bottom because the predicted chance cannot go much lower. [C099] A voice-mail plan lowers it: for the typical customer it acts on, from about 4% to about 2%. [C098] Its formula reproduces the part exactly on all 4000 training customers. [C103]

State units. This part reads only the state: it raises the chance of leaving in Rhode Island, Vermont and Tennessee, and lowers it in Virginia and Nevada. [C104, C105, C106, C107, C108] *Evidence.* This part reads only the state: it raises a typical customer's chance from about 1% to about 12% in Rhode Island, to about 3% in Vermont and to about 2% in Tennessee. [C104, C105, C106] It lowers the chance to below 1% in Virginia and in Nevada. [C107, C108] Its formula reproduces the part exactly on all 4000 training customers. [C116]

Worth checking against what is known. Worth checking: for customers in Rhode Island the fixed-weight inputs lower the chance to below 1% while the state units raise it to about 12%, so overall it is about 1%. [C061, C104, C025] Worth checking: for customers in Nevada the fixed-weight inputs raise the chance to about 4% while the state units lower it to below 1%, so overall it is about 2%. [C058, C108, C018] Worth checking: the state moves the chance by several points (Washington about 5%, Maine about 3%, Hawaii below 1%), based on 72 to 86 training customers per state. [C004, C006, C007] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C008, C002] Worth checking: the early rise with customer-service calls does not happen for customers in Nebraska. [C093]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C001	direction: international calling plan: no international plan to an international plan (whole model) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 18%	log-odds 2.74, OR 15.6; range across patients 2.74 to 2.74	3620, 380	train; test agrees (AUC gain 0.066 [0.035, 0.098], n=1000)	yes	clearly
C002	direction: voice-mail messages: 0 messages to 32 messages (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 6%	log-odds 1.61, OR 5.00; range across patients 1.61 to 1.61	4000	train; test agrees (AUC gain 0.014 [0.0002, 0.029], n=1000)	yes	clearly
C003	direction: total daytime minutes: 112 minutes to 249 minutes (whole model) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 6%	log-odds 1.30, OR 3.67; range across patients 0.49 to 1.59	4000	train; test agrees (AUC gain 0.074 [0.044, 0.105], n=1000)	yes	slightly
C004	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Washington (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 5%	log-odds 1.35, OR 3.86; range across patients 1.35 to 1.35	1693, 72	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C005	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 1.01, OR 2.73; range across patients 1.01 to 1.01	4000	train; test agrees (AUC gain 0.014 [0.003, 0.026], n=1000)	yes	slightly
C006	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Maine (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 3%	log-odds 0.71, OR 2.04; range across patients 0.71 to 0.71	1693, 86	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C007	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Hawaii (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to <1%	log-odds -2.50, OR 0.08; range across patients -2.50 to -2.50	1693, 73	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C008	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (whole model) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	log-odds -2.62, OR 0.07; range across patients -3.50 to -2.38	2939, 1061	train; test agrees (AUC gain 0.049 [0.028, 0.072], n=1000)	yes	slightly
C009	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts,	1% to <1%	log-odds -1.46, OR 0.23; range across patients -1.46 to -1.46	1693, 80	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C013	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Illinois (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds -0.94, OR 0.39; range across patients -0.94 to -0.94	1693, 68	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C014	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Pennsylvania (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds -0.77, OR 0.46; range across patients -0.77 to -0.77	1693, 56	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C015	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Virginia (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds -0.66, OR 0.52; range across patients -0.66 to -0.66	1693, 91	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C016	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Georgia (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds -0.64, OR 0.53; range across patients -0.64 to -0.64	1693, 60	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C017	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Mississippi (whole model) *shape: categorical; highest to lowest: Washington,*	1% to 2%	log-odds 0.39, OR 1.47; range across patients 0.39 to 0.39	1693, 77	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C025	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Rhode Island (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds -0.55, OR 0.58; range across patients -0.55 to -0.55	1693, 82	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C026	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Massachusetts (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	log-odds 0.35, OR 1.42; range across patients 0.35 to 0.35	1693, 86	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C027	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Nebraska (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds -0.81, OR 0.44; range across patients -0.83 to -0.53	1693, 72	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C028	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Florida (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds -0.44, OR 0.64; range across patients -0.44 to -0.44	1693, 78	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C029	direction: customer-service calls: 0 calls to 3 calls (whole model) *shape: accelerating; 99% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.30, OR 1.36; range across patients 0.31 to 0.31	4000	train; test agrees (AUC gain 0.106 [0.065, 0.145], n=1000)	yes	slightly
C030	direction: total daytime calls: 75 calls to 125 calls (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.32, OR 1.37; range across patients 0.32 to 0.32	4000	train; test disagrees (AUC gain -0.002 [-0.005, 0.001], n=1000)	yes	slightly; on the training data only
C031	direction: account length: 48 to 152 (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.28, OR 1.32; range across patients 0.28 to 0.28	4000	train; test disagrees (AUC gain -0.001 [-0.004, 0.001], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C037	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Arizona (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	1693, 73	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C038	direction: area code: area code 415 to area code other than area code 415 (whole model) *shape: categorical; highest to lowest: area code 415, area code other than area code 415*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	2016, 1984	train; test disagrees (AUC gain -7e-05 [-0.0004, 0.0003], n=1000)	yes	slightly; on the training data only
C039	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to California (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds -0.03, OR 0.97; range across patients 0.00 to 0.00	1693, 41	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C040	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Colorado (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds 0.11, OR 1.11; range across patients 0.00 to 0.22	1693, 78	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C041	magnitude: international calling plan: no international plan to an international plan, typical patient *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 18%	log-odds 2.74, OR 15.6	3620, 380	train; test agrees (AUC gain 0.066 [0.035, 0.098], n=1000)	yes	clearly
C042	magnitude: voice-mail messages: 0 messages to 32 messages, typical patient *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 6%	log-odds 1.61, OR 5.00	4000	train; test agrees (AUC gain 0.014 [0.0002, 0.029], n=1000)	yes	clearly
C043	magnitude: total daytime minutes: 112 minutes to 249 minutes, typical patient *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 6%	log-odds 1.59, OR 4.92	4000	train; test agrees (AUC gain 0.074 [0.044, 0.105], n=1000)	yes	slightly
C044	magnitude: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South	1% to 5%	log-odds 1.35, OR 3.86	1693, 72	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C049	null: model ignores total international charge (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C050	null: model ignores total international minutes (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C051	null: model ignores total night-time calls (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C052	null: model ignores total night-time charge (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C053	null: model ignores total night-time minutes (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C054	structure: made of linear_backbone, service_calls, day_usage_hinges, state_offsets; added		log-odds n/a, OR n/a	4000	train	yes	
C055	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 18%	log-odds 2.74, OR 15.6; range across patients 2.74 to 2.74	3620, 380	train; test agrees (AUC gain 0.066 [0.035, 0.098], n=1000)	yes	clearly
C056	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 6%	log-odds 1.61, OR 5.00; range across patients 1.61 to 1.61	4000	train; test agrees (AUC gain 0.014 [0.0002, 0.029], n=1000)	yes	clearly
C057	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Washington (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 5%	log-odds 1.35, OR 3.86; range across patients 1.35 to 1.35	2456, 72	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C058	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Nevada (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 4%	log-odds 1.05, OR 2.85; range across patients 1.05 to 1.05	2456, 77	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C059	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 1.01, OR 2.73; range across patients 1.01 to 1.01	4000	train; test agrees (AUC gain 0.014 [0.003, 0.026], n=1000)	yes	slightly
C060	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Maine (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 3%	log-odds 0.71, OR 2.04; range across patients 0.71 to 0.71	2456, 86	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C061	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Rhode Island (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to <1%	log-odds -2.85, OR 0.06; range across patients -2.85 to -2.85	2456, 82	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C062	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Hawaii (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to <1%	log-odds -2.50, OR 0.08; range across patients -2.50 to -2.50	2456, 73	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C063	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	log-odds -2.38, OR 0.09; range across patients -2.38 to -2.38	2939, 1061	train; test agrees (AUC gain 0.038 [0.017, 0.061], n=1000)	yes	slightly
C064	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Vermont (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to <1%	log-odds -2.22, OR 0.11; range across patients -2.22 to -2.22	2456, 80	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C065	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to South Carolina (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 2%	log-odds 0.54, OR 1.71; range across patients 0.54 to 0.54	2456, 66	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C066	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Alaska (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	log-odds -0.97, OR 0.38; range across patients -0.97 to -0.97	2456, 56	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C067	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada,	1% to 1%	log-odds -0.94, OR 0.39; range across patients -0.94 to -0.94	2456, 68	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C073	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Nebraska (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	log-odds -0.53, OR 0.59; range across patients -0.53 to -0.53	2456, 72	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C074	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.32, OR 1.37; range across patients 0.32 to 0.32	4000	train; test disagrees (AUC gain -0.002 [-0.005, 0.001], n=1000)	yes	slightly; on the training data only
C075	direction: account length: 48 to 152 (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.28, OR 1.32; range across patients 0.28 to 0.28	4000	train; test disagrees (AUC gain -0.001 [-0.004, 0.001], n=1000)	yes	slightly; on the training data only
C076	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Oregon (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 2%	log-odds 0.23, OR 1.26; range across patients 0.23 to 0.23	2456, 84	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C077	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Florida (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	log-odds -0.30, OR 0.74; range across patients -0.30 to -0.30	2456, 78	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C078	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to North Carolina (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	log-odds -0.13, OR 0.87; range across patients -0.13 to -0.13	2456, 74	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C079	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to North Dakota (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	log-odds 0.06, OR 1.07; range across patients 0.06 to 0.06	2456, 68	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C080	direction: state: state	1% to 1%	log-odds 0.03, OR 1.03;	2456,	train; test disagrees	yes	slightly; on

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C085	direction: state: state other than Nebraska, California, Colorado to Nebraska (through this part) *shape: categorical; highest to lowest: Nebraska, California, Colorado, state other than Nebraska, California, Colorado*	1% to 1%	log-odds -0.29, OR 0.75; range across patients -0.31 to 0.00	3809, 72	train; test unclear (AUC gain 0.0001 [-0.0007, 0.002], n=1000)	yes	slightly; on the training data only
C086	direction: state: state other than Nebraska, California, Colorado to California (through this part) *shape: categorical; highest to lowest: Nebraska, California, Colorado, state other than Nebraska, California, Colorado*	1% to 1%	log-odds -0.03, OR 0.97; range across patients 0.00 to 0.00	3809, 41	train; test unclear (AUC gain 0.0001 [-0.0007, 0.002], n=1000)	yes	slightly; on the training data only
C087	direction: state: state other than Nebraska, California, Colorado to Colorado (through this part) *shape: categorical; highest to lowest: Nebraska, California, Colorado, state other than Nebraska, California, Colorado*	1% to 1%	log-odds 0.11, OR 1.11; range across patients 0.00 to 0.22	3809, 78	train; test unclear (AUC gain 0.0001 [-0.0007, 0.002], n=1000)	yes	slightly; on the training data only
C088	interaction: effect of customer-service calls (0 calls to 3 calls) by state: Nebraska minus state other than Nebraska, California, Colorado; stronger in state other than Nebraska, California, Colorado *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	Nebraska: 1% to 1%; state other than Nebraska, California, Colorado: 1% to 2%	log-odds difference -0.31, OR 0.74	72, 3809	train; test agrees (AUC gain 0.0001 [-0.0007, 0.002], n=1000); group difference on test -0.31 [-0.31, -0.31]	yes	slightly
C089	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 99% of patients same; verified; risk saturates near 0%*	1% to 10% (typical patient with a fold)	fold 2.73 calls (IQR 2.73 to 2.73); past it log-odds 2.06, OR 7.84; range across patients 2.08 to 2.08	4000, 831	train; test agrees (AUC gain 0.106 [0.065, 0.145], n=1000)	yes	clearly
C090	structure: made of csc_late_chain, csc_early_hinge; added		log-odds n/a, OR n/a	4000	train	yes	
C091	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 2%	log-odds 0.30, OR 1.35; range across patients 0.31 to 0.31	4000	train; test agrees (AUC gain 0.082 [0.050, 0.114], n=1000)	yes	slightly
C092	direction: state: state other than Nebraska to Nebraska (through this part) *shape: categorical; highest to lowest: state other than Nebraska, Nebraska*	2% to 1% (typical patient this part acts on)	log-odds -0.29, OR 0.75; range across patients -0.31 to 0.00	3928, 72	train; test unclear (AUC gain 8e-05 [-0.0002, 0.0003], n=1000)	yes	slightly; on the training data only
C093	interaction: effect of customer-service calls (0 calls to 3 calls) by state: Nebraska minus state other than Nebraska; stronger in state other than Nebraska *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	Nebraska: 1% to 1%; state other than Nebraska: 1% to 2%	log-odds difference -0.31, OR 0.73	72, 3928	train; test agrees (AUC gain 8e-05 [-0.0002, 0.0003], n=1000); group difference on test -0.31 [-0.31, -0.31]	yes	slightly
C094	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 10% (typical patient with a fold)	fold 2.73 calls (IQR 2.73 to 2.73); past it log-odds 2.08, OR 8.00; range across patients 2.08 to 2.08	3928, 830	train; test agrees (AUC gain 0.082 [0.050, 0.114], n=1000)	yes	clearly
C095	subgroup: who this part acts on: 830 of 4000 patients	1% to 2% (typical patient this part acts on (risk without the part, then with it))	share 0.21; log-odds 1.36, OR 3.88; range across patients 0.31 to 3.85	830, 3170	train; test agrees (AUC gain 0.083 [0.051, 0.116], n=1000)	yes	slightly
C096	structure: its formula reproduces it on 4000 training patients (largest difference 8.88178e-16)		log-odds 9e-16, OR n/a	4000	train	yes	

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C097	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 6%	log-odds 1.30, OR 3.67; range across patients 0.49 to 1.59	4000	train; test agrees (AUC gain 0.074 [0.044, 0.105], n=1000)	yes	slightly
C098	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	4% to 2% (typical patient this part acts on)	log-odds -0.24, OR 0.79; range across patients -1.12 to 0.00	2939, 1061	train; test agrees (AUC gain 0.007 [0.003, 0.011], n=1000)	yes	slightly
C099	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by voice-mail plan: a voice-mail plan minus no voice-mail plan; stronger in no voice-mail plan *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	a voice-mail plan: <1% to <1%; no voice-mail plan: 1% to 6%	log-odds difference -1.11, OR 0.33	1061, 2939	train; test agrees (AUC gain 0.007 [0.003, 0.011], n=1000); group difference on test -1.11 [-1.11, -1.11]	yes	slightly
C100	threshold: total daytime minutes: fold at about 209 minutes (starts) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 16% (typical patient with a fold)	fold 209 minutes (IQR 209 to 224); past it log-odds 2.24, OR 9.41; range across patients 1.15 to 2.63	4000, 1074	train; test agrees (AUC gain 0.074 [0.044, 0.105], n=1000)	yes	clearly
C101	threshold: total daytime charge: fold at about 51.68 US dollars (starts) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient with a fold)	fold 51.7 US dollars (IQR 51.7 to 51.7); past it log-odds 0.00, OR 1.00; range across patients 0.00 to 0.00	4000, 41	train; test unclear (AUC gain 3e-05 [-3e-05, 0.0001], n=1000)	no	slightly; on the training data only
C102	subgroup: who this part acts on: 1074 of 4000 patients	1% to 4% (typical patient this part acts on (risk without the part, then with it))	share 0.27; log-odds 1.28, OR 3.61; range across patients 0.12 to 3.02	1074, 2926	train; test agrees (AUC gain 0.080 [0.049, 0.113], n=1000)	yes	slightly
C103	structure: its formula reproduces it on 4000 training patients (largest difference 2.22045e-16)		log-odds 2e-16, OR n/a	4000	train	yes	
C104	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Rhode Island (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 12%	log-odds 2.30, OR 9.97; range across patients 2.30 to 2.30	3039, 82	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	clearly
C105	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Vermont (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 3%	log-odds 0.77, OR 2.16; range across patients 0.77 to 0.77	3039, 80	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C106	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Tennessee (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	log-odds 0.54, OR 1.72; range across patients 0.54 to 0.54	3039, 68	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C107	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Virginia (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 1%	log-odds -0.66, OR 0.52; range across patients -0.66 to -0.66	3039, 91	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C108	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia,	1% to 1%	log-odds -0.66, OR 0.52; range across patients -0.66 to -0.66	3039, 77	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C109	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Mississippi (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	log-odds 0.39, OR 1.47; range across patients 0.39 to 0.39	3039, 77	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C110	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Kansas (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	log-odds 0.38, OR 1.46; range across patients 0.38 to 0.38	3039, 77	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C111	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Montana (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	log-odds 0.38, OR 1.46; range across patients 0.38 to 0.38	3039, 75	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C112	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to New York (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	log-odds 0.38, OR 1.46; range across patients 0.38 to 0.38	3039, 93	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C113	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Indiana (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	log-odds 0.37, OR 1.45; range across patients 0.37 to 0.37	3039, 82	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C114	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Delaware (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 1%	log-odds -0.20, OR 0.82; range across patients -0.20 to -0.20	3039, 81	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C115	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Florida (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 1%	log-odds -0.14, OR 0.87; range across patients -0.14 to -0.14	3039, 78	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C116	structure: its formula reproduces it on 4000 training patients (largest difference 1.11022e-16)		log-odds 1e-16, OR n/a	4000	train	yes	
C117	direction: state: state other than California, Colorado to Colorado (through this part) *shape: categorical; highest to lowest: Colorado, state*	52% to 75% (typical patient this part acts on)	log-odds 0.11, OR 1.11; range across patients 0.00 to 0.22	3881, 78	train; test unclear (AUC gain 2e-05 [-0.0009, 0.001], n=1000)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C121	threshold: customer-service calls: fold at about 5 calls (starts) *shape: hinge at 4 calls; 98% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient with a fold)	fold 4.21 calls (IQR 4.21 to 4.21); past it log-odds 0.02, OR 1.02; range across patients 0.00 to 0.00	3959, 134	train; test unclear (AUC gain 0.0003 [-0.0007, 0.002], n=1000)	no	slightly; on the training data only
C122	subgroup: who this part acts on: 134 of 4000 patients	39% to 75% (typical patient this part acts on (risk without the part, then with it))	share 0.03; log-odds 0.85, OR 2.35; range across patients 0.53 to 1.88	134, 3866	train; test unclear (AUC gain 0.0003 [-0.0008, 0.002], n=1000)	yes	on the training data only
C123	structure: its formula reproduces it on 4000 training patients (largest difference 2.22045e-16)		log-odds 2e-16, OR n/a	4000	train	yes	

Not supported by the model.

- C101: threshold claim about total daytime charge in part ann_85: not supported by the model (the effect past the fold is too small or not consistent across patients).
- C121: threshold claim about number of calls to customer service in part ann_83: not supported by the model (the effect past the fold is too small or not consistent across patients).

1.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (30):** `-10` (state:17), `-13` (state:2), `-14` (state:20), `-16` (state:22), `-17` (state:23), `-18` (state:24), `-2` (state:1), `-25` (state:30), `-26` (state:31), `-27` (state:32), `-30` (state:35), `-37` (state:41), `-39` (state:43), `-44` (state:48), `-45` (state:49), `-47` (state:50), `-48` (state:6), `-49` (state:7), `-5` (state:12), `-53` (areaCode:408), `-55` (areaCode:510), `-6` (state:13), `-62` (totalEveMinutes), `-63` (totalEveCalls), `-65` (totalNightMinutes), `-66` (totalNightCalls), `-67` (totalNightCharge), `-68` (totalIntlMinutes), `-69` (totalIntlCalls), `-70` (totalIntlCharge).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

1.3 Annotation hierarchy

The hierarchy has 1 root; the global root is churn_logit. Tags (A1, A2, . . .) are used in the figures.

▷ A1: churn_logit (composition, section 1.6.1)
  ▷ A2: linear_backbone (leaf, section 1.6.2)
  ▷ A3: day_usage_hinges (leaf, section 1.6.3)
  ▷ A4: state_offsets (leaf, section 1.6.4)
  ▷ A5: service_calls (composition, section 1.6.5)
    ▷ A6: csc_late_chain (leaf, section 1.6.6)
    ▷ A7: csc_early_hinge (leaf, section 1.6.7)

1.4 Network figures

The explained model state has more than 60 non-Null nodes, so the collapsed views are shown; the full view is in `churn100-s4-c3-fig-full.tex`.

1.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 4000 training rows is 4.44×10^{-16} (tolerance 1.00×10^{-9}).

- #71: split `-71` into `-71_a`, `-71_b`, `-71_c`.
- #72: split `-60` into `-60_a`, `-60_b`.
- #73: split `-57` into `-57_a`, `-57_b`.
- #74: split `-51` into `-51_a`, `-51_b`.
- #75: split `-42` into `-42_a`, `-42_b`.
- #76: split `-34` into `-34_a`, `-34_b`.
- #77: split `-28` into `-28_a`, `-28_b`.
- #78: split `-23` into `-23_a`, `-23_b`.
- #79: consolidate `-71_a`, `-71_c`.
- #80: identity node `identity_lin` before `0` (28 connection(s)).
- #81: identity node `identity_svc` before `0` (3 connection(s)).

1.6 Annotations

1.6.1 A1: churn_logit

Kind: Composition annotation (depth 0); children A2 (section 1.6.2), A3 (section 1.6.3), A4 (section 1.6.4), A5 (section 1.6.5).

- **Entry nodes:** `-70` (totalIntlCharge), `-69` (totalIntlCalls), `-68` (totalIntlMinutes), `-67` (totalNightCharge), `-66` (totalNightCalls), `-65` (totalNightMinutes), `-64` (totalEveCharge), `-63` (totalEveCalls), `-62` (totalEveMinutes), `-61` (totalDayCharge), `-59` (totalDayMinutes), `-58` (numberVmailMessages), `-56` (internationalPlan), `-55` (areaCode:510), `-54` (areaCode:415), `-53` (areaCode:408), `-52` (accountLength), `-50` (state:8), `-49` (state:7), `-48` (state:6), `-47` (state:50), `-46` (state:5), `-45` (state:49), `-44` (state:48), `-43` (state:47), `-41` (state:45), `-40` (state:44), `-39` (state:43), `-38` (state:42), `-37` (state:41), `-36` (state:40), `-35` (state:4), `-33` (state:38), `-32` (state:37), `-31` (state:36), `-30` (state:35), `-29` (state:34), `-27` (state:32), `-26` (state:31), `-25` (state:30), `-24` (state:3), `-22` (state:28), `-21` (state:27), `-20` (state:26), `-19` (state:25), `-18` (state:24), `-17` (state:23), `-16` (state:22), `-15` (state:21), `-14` (state:20), `-13` (state:2), `-12` (state:19), `-11` (state:18), `-10` (state:17), `-9` (state:16), `-8` (state:15), `-7` (state:14), `-6` (state:13), `-5` (state:12), `-4` (state:11), ... (19 more; see the JSON record).
- **Exit nodes:** `0`.
- **Internal nodes:** none.
- **Edges (weight):** `1295`→`0` (-1.11), `1390`→`0` (-1.17), `3118`→`0` (1.48), `4000`→`0` (1.5), `5533`→`0` (1.98), `5786`→`0` (-0.557), `7872`→`0` (-1.25), `9954`→`0` (0.056), `10131`→`0` (1.89), `11361`→`0` (0.0653), `12690`→`0` (-0.0283), `13068`→`0` (-0.0199), `15362`→`0` (0.0582), `15490`→`0` (0.0266), `15904`→`0` (0.0449), `17102`→`0` (-0.13), `identity_lin`→`0` (1), `identity_svc`→`0` (1).

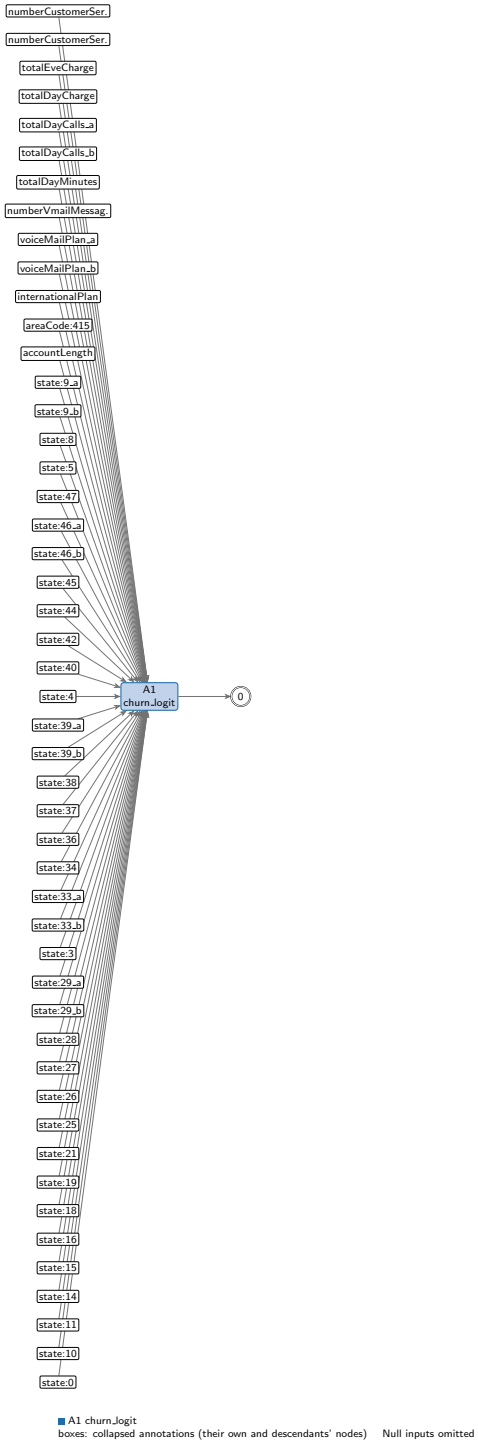


Figure 1: churn100-s4-c3: collapsed view at hierarchy level 0 (fully recomposed).

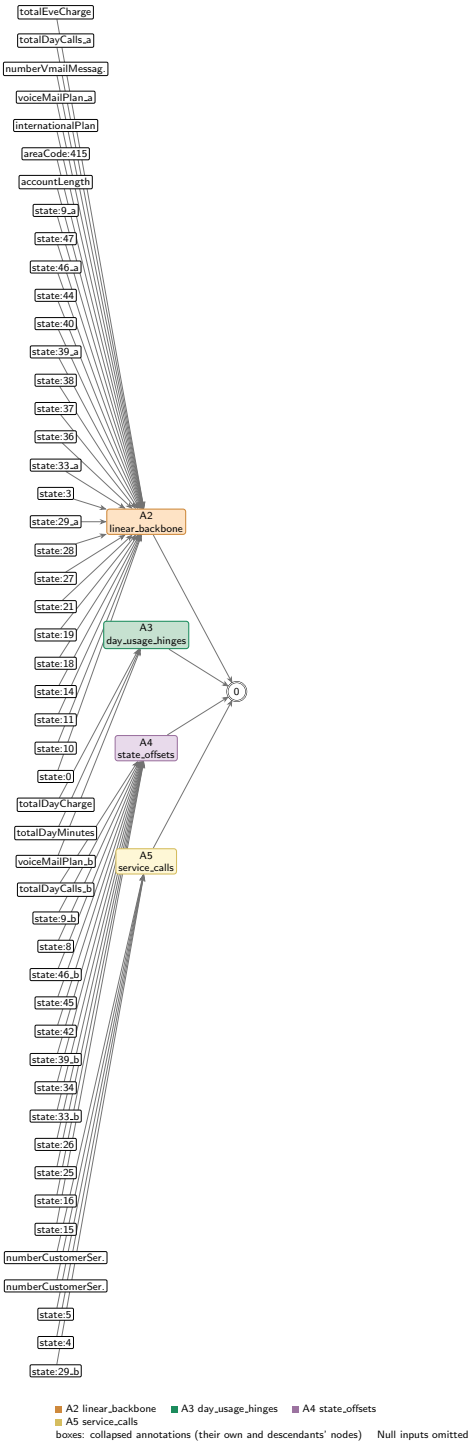


Figure 2: churn100-s4-c3: collapsed view at hierarchy level 1.

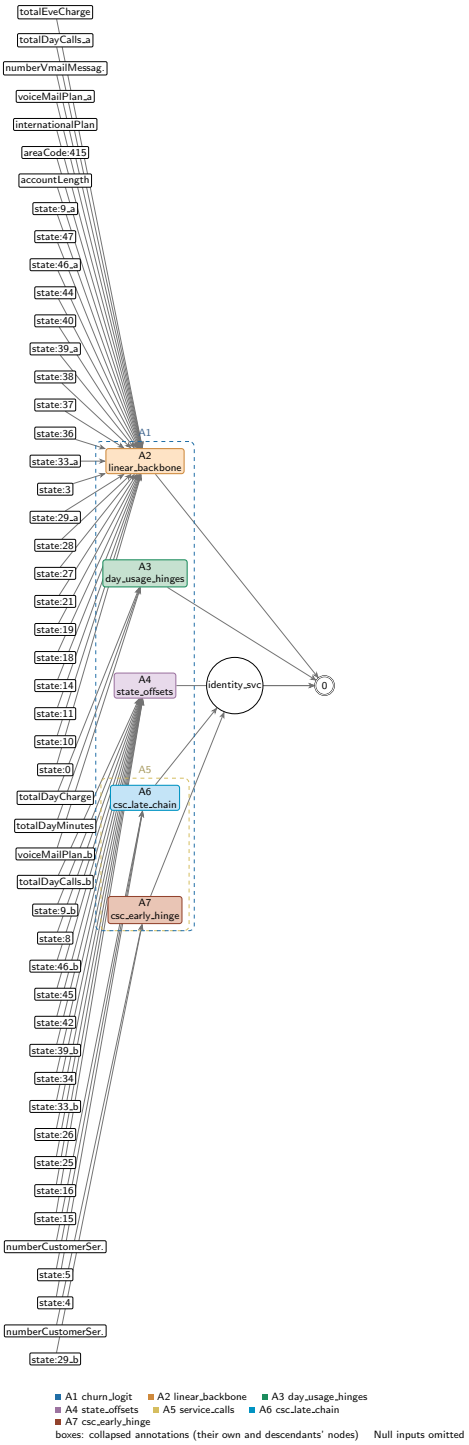


Figure 3: churn100-s4-c3: collapsed view at hierarchy level 2.

- **With its descendants:** 103 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 102 candidate nodes.

Hypothesis. Root: logit P(churn) = -1.35 + linear_backbone (identity_lin) + service_calls (identity_svc) + 1.48*3118 + 1.50*4000 + 1.89*10131 (day_usage_hinges) + the state_offsets units (-1.11*1295 - 1.25*7872 - 1.17*1390 - 0.56*5786 + small terms; 5533 is never active), purely additive in the logit; the sigmoid is the only coupling. 30 inputs are unconnected (international minutes, calls and charge, night minutes/charge/calls, evening minutes and calls, area codes 408/510 and 20 state indicators) and have zero effect. Train AUC 0.8884, accuracy 0.8795; the mean predicted churn is 0.082 against a base rate of 0.1415 (the model ranks well but under-predicts). Performance sources (train): all children replaced by least-squares linear fits gives AUC 0.8352 (an L2 logistic regression refit on all 71 inputs gets 0.8433); restoring day_usage_hinges alone gives 0.8713, service_calls alone 0.8657, both 0.8884; the state units restore nothing (exactly linear). So the gain over linear (+0.053) comes from the day-usage and service-call thresholds, which are complementary. Rare-category switches: state:11 -2.50 logit (73 rows, empirical -1.35); state:46 -1.46 net (80 rows, -0.91); state:47 +1.35 (72 rows, +0.92); state:0 -0.97 (56, -0.76); state:14 -0.94 (68, -0.53); state:29 -0.53 direct plus switching off the early service-call hinge (mean -0.78, down to -6.15 for many-call customers; 72 rows, -0.79); state:39 -0.55 net (82, -0.93: direct -2.85 mostly cancelled by the state units); state:33 +0.38 net (direct +1.05, unit -0.68); state:18 +0.36 (65 rows, empirical -0.48: opposite sign); state:4 switches off the late chain (41 rows, empirical +0.80).

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma\big(z_0\big)\\
z_0 &= -1.35 + 1.0\,\mathrm{linear_backbone}\big(x_{state:0}, x_{state:18},\\
&\quad x_{state:19}, x_{state:21}, x_{state:27}, x_{state:28}, x_{state:29_a}, x_{state:3},\\
&\quad x_{state:33_a}, x_{state:10}, x_{state:36}, x_{state:37}, x_{state:38}, x_{state:39_a},\\
&\quad x_{state:40}, x_{state:11}, x_{state:44}, x_{state:46_a}, x_{state:47}, x_{state:9_a},\\
&\quad x_{accountLength}, x_{areaCode:415}, x_{internationalPlan}, x_{voiceMailPlan_a},\\
&\quad x_{numberVmailMessages}, x_{totalDayCalls_a}, x_{totalEveCharge}, x_{state:14}\big)\\
&\quad + 1.0\,\mathrm{service_calls}\big(x_{numberCustomerServiceCalls_ac}, x_{state:5},\\
&\quad x_{state:4}, x_{state:29_b}, x_{numberCustomerServiceCalls_b}\big)\\
&\quad + 1.48\,\mathrm{day_usage_hinges_0}\big(x_{voiceMailPlan_b}, x_{totalDayMinutes},\\
&\quad x_{totalDayCharge}\big) + 1.5\,\mathrm{day_usage_hinges_1}\big(x_{voiceMailPlan_b},\\
&\quad x_{totalDayMinutes}, x_{totalDayCharge}\big)\\
&\quad + 0.0266\,\mathrm{state_offsets_8}\big(x_{totalDayCalls_b}, x_{state:33_b},\\
&\quad x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b},\\
&\quad x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big)\\
&\quad + 0.056\,\mathrm{state_offsets_3}\big(x_{totalDayCalls_b}, x_{state:33_b},\\
&\quad x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b},\\
&\quad x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big)\\
&\quad + 1.89\,\mathrm{day_usage_hinges_2}\big(x_{voiceMailPlan_b}, x_{totalDayMinutes},\\
&\quad x_{totalDayCharge}\big) + 0.0653\,\mathrm{state_offsets_4}\big(x_{totalDayCalls_b},\\
&\quad x_{state:33_b}, x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16},\\
&\quad x_{state:9_b}, x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big)\\
&\quad + 0.0449\,\mathrm{state_offsets_9}\big(x_{totalDayCalls_b}, x_{state:33_b},\\
&\quad x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b},\\
&\quad x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big)\\
&\quad + 0.0582\,\mathrm{state_offsets_7}\big(x_{totalDayCalls_b}, x_{state:33_b},\\
&\quad x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b},
\end{aligned}
$$

$$
\begin{aligned}
&x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&+ 1.98\,\text{state_offsets_12}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&- 0.0199\,\text{state_offsets_6}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&- 1.17\,\text{state_offsets_2}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&- 0.557\,\text{state_offsets_11}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&- 1.25\,\text{state_offsets_1}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&- 0.13\,\text{state_offsets_10}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&- 1.11\,\text{state_offsets_0}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&- 0.0283\,\text{state_offsets_5}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big)
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 &= \sigma\big(z_0\big) \\
z_0 &= -1.35 + 0.0337x_{state:37} + 0.144x_{state:33_a} + 0.679x_{numberVmailMessages} \\
&\quad + 1.48\,\text{relu}\big(-0.836 + 0.365x_{totalDayCharge}\big) \\
&\quad + 1.5\,\text{relu}\big(-0.565 + 0.7x_{totalDayMinutes}\big) \\
&\quad + 0.0266\,\text{relu}\big(0.361 + 0.984x_{state:42}\big) \\
&\quad + 0.056\,\text{relu}\big(-0.38 + 1.01x_{state:15}\big) \\
&\quad + 1.88\,\text{relu}\big(-0.615 + 0.52x_{numberCustomerServiceCalls_b} \\
&\quad - 1.12x_{state:29_b}\big) + 1.89\,\text{relu}\big(-0.625 \\
&\quad + 0.789x_{totalDayMinutes} - 0.347x_{voiceMailPlan_b}\big) \\
&\quad + 0.0653\,\text{relu}\big(1.36 + 1.65x_{state:46_b}\big) + 0.0685x_{state:40} \\
&\quad + 0.00461x_{state:36} + 0.0449\,\text{relu}\big(-0.0356 \\
&\quad + 0.993\,\text{relu}\big(0.361 + 0.984x_{state:42}\big)\big) + 0.18x_{state:47} \\
&\quad + 0.398x_{totalEveCharge} + 0.103x_{state:21} + 1.71\,\text{relu}\big(-0.557 \\
&\quad + 0.256\,\text{relu}\big(0.476 + 0.74x_{numberCustomerServiceCalls_ac}\big) \\
&\quad + 0.327x_{numberCustomerServiceCalls_ac} - 1.39\,\text{relu}\big(0.219 \\
&\quad + 0.414x_{state:4} - 1.86x_{state:5}\big)\big) + 0.126x_{totalDayCalls_a} \\
&\quad + 0.00838x_{state:28} + 0.00889x_{areaCode:415} + 0.0459x_{state:18} \\
&\quad + 0.805x_{internationalPlan} + 0.0513x_{state:19} + 0.106x_{accountLength} \\
&\quad + 0.0534x_{state:44} + 1.73\,\text{relu}\big(-0.0961
\end{aligned}
$$

$$
\begin{aligned}
&+ 1.49\,\mathrm{relu}\big(-0.615 + 0.52x_{numberCustomerServiceCalls_b} \\
&- 1.12x_{state:29_b}\big)\big) + 0.0582\,\mathrm{relu}\big(-0.581 \\
&+ 0.999x_{state:16}\big) + 1.98\,\mathrm{relu}\big(-1.54 \\
&+ 0.655\,\mathrm{relu}\big(0.383 + 0.163x_{state:33_b} - 1.5x_{state:26} \\
&- 0.732x_{state:34}\big) - 0.00543x_{totalDayCalls_b}\big) \\
&- 0.0199\,\mathrm{relu}\big(-0.372 + 1.02x_{state:9_b}\big) \\
&- 1.05x_{voiceMailPlan_a} - 0.335x_{state:11} - 1.17\,\mathrm{relu}\big(0.252 \\
&- 0.574x_{state:25}\big) - 0.404x_{state:39_a} - 0.557\,\mathrm{relu}\big(0.383 \\
&+ 0.163x_{state:33_b} - 1.5x_{state:26} - 0.732x_{state:34}\big) \\
&- 1.25\,\mathrm{relu}\big(0.0571 + 1.02\,\mathrm{relu}\big(0.78 \\
&- 1.28x_{state:39_b}\big)\big) - 0.311x_{state:46_a} - 0.0781x_{state:10} \\
&- 0.0904x_{state:38} - 0.00327x_{state:3} - 0.114x_{state:0} \\
&- 0.13\,\mathrm{relu}\big(-1.42 + 0.995x_{state:45}\big) \\
&- 1.11\,\mathrm{relu}\big(0.78 - 1.28x_{state:39_b}\big) - 0.07x_{state:29_a} \\
&- 0.0181x_{state:27} - 0.0419x_{state:9_a} - 0.0283\,\mathrm{relu}\big(-0.0337 \\
&+ 1.0x_{state:8}\big) - 0.122x_{state:14}
\end{aligned}
$$

where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00303, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Root: logit P(churn) = -1.35 + linear_backbone (identity_lin) + service_calls (identity_svc) + 1.48*3118 + 1.50*4000 + 1.89*10131 (day_usage_hinges) + the state_offsets units (-1.11*1295 - 1.25*7872 - 1.17*1390 - 0.56*5786 + small terms; 5533 is never active), purely additive in the logit; the sigmoid is the only coupling. 30 inputs are unconnected (international minutes, calls and charge, night m

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.2 A2: linear_backbone

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-1` (state:0), `-11` (state:18), `-12` (state:19), `-15` (state:21), `-21` (state:27), `-22` (state:28), `-23_a` (state:29_a), `-24` (state:3), `-28_a` (state:33_a), `-3` (state:10), `-31` (state:36), `-32` (state:37), `-33` (state:38), `-34_a` (state:39_a), `-36` (state:40), `-4` (state:11), `-40` (state:44), `-42_a` (state:46_a), `-43` (state:47), `-51_a` (state:9_a), `-52` (accountLength), `-54` (areaCode:415), `-56` (internationalPlan), `-57_a` (voiceMailPlan_a), `-58` (numberVmailMessages), `-60_a` (totalDayCalls_a), `-64` (totalEveCharge), `-7` (state:14).
- **Exit nodes:** `identity_lin`.
- **Internal nodes:** none.
- **Edges (weight):** `-1`→`identity_lin` (-0.114), `-3`→`identity_lin` (-0.0781), `-4`→`identity_lin` (-0.335), `-7`→`identity_lin` (-0.122), `-11`→`identity_lin` (0.0459), `-12`→`identity_lin` (0.0513), `-15`→`identity_lin` (0.103), `-21`→`identity_lin` (-0.0181), `-22`→`identity_lin` (0.00838), `-24`→`identity_lin` (-0.00327), `-31`→`identity_lin` (0.00461), `-32`→`identity_lin` (0.0337), `-33`→`identity_lin` (-0.0904), `-36`→`identity_lin` (0.0685), `-40`→`identity_lin` (0.0534), `-43`→`identity_lin` (0.18), `-52`→`identity_lin` (0.106), `-54`→`identity_lin` (0.00889), `-56`→`identity_`

lin (0.805), -58→identity_lin (0.679), -64→identity_lin (0.398), -60_a→identity_lin (0.126), -57_a→identity_lin (-1.05), -51_a→identity_lin (-0.0419), -42_a→identity_lin (-0.311), -34_a→identity_lin (-0.404), -28_a→identity_lin (0.144), -23_a→identity_lin (-0.07).

- **Covers** (coverage, own and as part of the hierarchy): 28 candidate nodes.

Hypothesis. Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 28 inputs: voicemail plan -1.05 per SD (-2.61 logit for having a plan), international plan +0.80 (+2.75 logit for having one), voicemail messages +0.68 per SD (offsets most of the plan effect for heavy message users), evening charge +0.40, day calls +0.13, account length +0.11, and state offsets: state:39 -0.40 per SD (-2.85 logit, but state_offsets gives back +2.30), state:11 -0.33 (-2.50), state:46 -0.31 (-2.22, of which state_offsets gives back +0.77), state:47 +0.18 (+1.35), state:33 +0.14 (+1.05, of which state_offsets takes back -0.68), state:0 -0.97, state:14 -0.94, state:38 -0.77, state:21 +0.71, state:10 -0.64, state:40 +0.54, state:29 -0.53, then smaller states and area code 415 (+0.01). Day minutes, day charge and customer-service calls have NO direct edge: they act only through hinges. Evidence: closed form exact (identity); train sensitivities equal the weights (voicemail plan 1.05, international plan 0.80, voicemail messages 0.68, state:39 0.40, evening charge 0.40).

Closed form:

$$
\begin{aligned}
y_0 = {} & 0.0337x_{state:37} + 0.144x_{state:33_a} + 0.679x_{numberVmailMessages} \\
& + 0.0685x_{state:40} + 0.00461x_{state:36} + 0.18x_{state:47} + 0.398x_{totalEveCharge} \\
& + 0.103x_{state:21} + 0.126x_{totalDayCalls_a} + 0.00838x_{state:28} \\
& + 0.00889x_{areaCode:415} + 0.0459x_{state:18} + 0.805x_{internationalPlan} \\
& + 0.0513x_{state:19} + 0.106x_{accountLength} + 0.0534x_{state:44} \\
& - 1.05x_{voiceMailPlan_a} - 0.335x_{state:11} - 0.404x_{state:39_a} \\
& - 0.311x_{state:46_a} - 0.0781x_{state:10} - 0.0904x_{state:38} - 0.00327x_{state:3} \\
& - 0.114x_{state:0} - 0.07x_{state:29_a} - 0.0181x_{state:27} - 0.0419x_{state:9_a} \\
& - 0.122x_{state:14}
\end{aligned}
$$

where y_0 is node identity_lin; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00552, and the exact function by at most 1.78×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, analytical_formula; data split: train. Closed form from get_formula (force).
- EMPIRICAL, empirical_viz; data split: train. Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 28 inputs: voicemail plan -1.05 per SD (-2.61 logit for having a plan), international plan +0.80 (+2.75 logit for having one), voicemail messages +0.68 per SD (offsets most of the plan effect for heavy message users), evening charge +0.40, day calls +0.13, account length +0.11, and state offsets: stat

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.3 A3: day_usage_hinges

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** -57_b (voiceMailPlan_b), -59 (totalDayMinutes), -61 (totalDayCharge).
- **Exit nodes:** 3118, 4000, 10131.
- **Internal nodes:** none.

- **Edges (weight):** -59→4000 (0.7), -61→3118 (0.364), -59→10131 (0.789), -57_b→10131 (-0.347).
- **Covers** (coverage, own and as part of the hierarchy): 3 candidate nodes.

Hypothesis. 10131 = relu(-0.63 + 0.79 dayMin - 0.35 voiceMailPlan), logit weight +1.89; 4000 = relu(-0.56 + 0.70 dayMin), weight +1.50; 3118 = relu(-0.84 + 0.36 dayCharge), weight +1.48. Day minutes and day charge are collinear (r = 1.0000), so all three are hinges in day usage. Hypothesis: day usage has no effect below about 200 minutes, then raises churn risk steeply in three stages: 10131 switches on at about 209 day minutes without a voicemail plan (about 263 with one), 4000 at about 224 minutes, and 3118 at a day charge of about 51.7 (about 304 minutes; on for only 1% of customers). Evidence (train): combined logit contribution 0 below 200 minutes, +0.05 at 200-220, +0.51 at 220-240, +1.31 at 240-260, +2.19 at 260-280, +3.10 at 280-300, +4.69 above 300 (voicemail-plan holders about 1.1-1.7 lower from 240 minutes up); churn by band 0.06/0.14/0.26/0.46/0.58/0.79 against model 0.04/0.08/0.15/0.24/0.37/0.64 (right shape, under-predicted, as is the whole model). Sensitivity: 10131 day minutes 0.18, voicemail plan 0.08; 4000 day minutes 0.14; 3118 day charge 0.004. Mean ablation AUC 0.8083 (the largest single drop); linear replacement 0.8657 (R2 0.47/0.44/0.05); restored alone onto all-linear 0.8713 (all-linear 0.8352).

Closed form:

$$
\begin{aligned}
y_0 &= \mathrm{relu}\big(-0.836 + 0.365 x_{totalDayCharge}\big) \\
y_1 &= \mathrm{relu}\big(-0.565 + 0.7 x_{totalDayMinutes}\big) \\
y_2 &= \mathrm{relu}\big(-0.625 + 0.789 x_{totalDayMinutes} \\
&\qquad - 0.347 x_{voiceMailPlan_b}\big)
\end{aligned}
$$

where y_0 is node 3118; y_1 is node 4000; y_2 is node 10131; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00207, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, analytical_formula; data split: train. Closed form from get_formula (force).
- EMPIRICAL, ablation_study; data split: train. 10131 = relu(-0.63 + 0.79 dayMin - 0.35 voiceMailPlan), logit weight +1.89; 4000 = relu(-0.56 + 0.70 dayMin), weight +1.50; 3118 = relu(-0.84 + 0.36 dayCharge), weight +1.48. Day minutes and day charge are collinear (r = 1.0000), so all three are hinges in day usage. Hypothesis: day usage has no effect below about 200 minutes, then raises churn risk steeply in three stages: 10131 switches on at ab

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1.6.4 A4: state_offsets

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** -60_b (totalDayCalls_b), -28_b (state:33_b), -29 (state:34), -20 (state:26), -41 (state:45), -38 (state:42), -9 (state:16), -51_b (state:9_b), -50 (state:8), -42_b (state:46_b), -8 (state:15), -19 (state:25), -34_b (state:39_b).
- **Exit nodes:** 1295, 7872, 1390, 9954, 11361, 12690, 13068, 15362, 15490, 15904, 17102, 5786, 5533.
- **Internal nodes:** none.
- **Edges (weight):** -8→9954 (1.01), -19→1390 (-0.574), -20→5786 (-1.5), -29→5786 (-0.732), -9→15362 (0.999), -38→15490 (0.984), -41→17102 (0.995), -50→12690 (1), 1295→7872 (1.02), 5786→5533 (0.655), 15490→15904 (0.993), -60_b→5533 (-0.00543), -51_b→13068 (1.02), -42_b→11361 (1.65), -34_b→1295 (-1.28), -28_b→5786 (0.163).

- **Covers** (coverage, own and as part of the hierarchy): 13 candidate nodes.

Hypothesis. Thirteen units, each reading only one-hot state indicators, so each is exactly an offset per state (least-squares R2 = 1.0): 1295 -> 7872 (state:39; weights -1.11, -1.25: a constant -2.32 logit for everyone except state:39, where it is switched off, +2.30, against state:39's direct -2.85; net -0.55 vs empirical log-odds offset -0.93, 82 customers), 11361 (state:46, +0.77 against direct -2.22), 5786 (state:26/33/34: +0.38 for state:26 and state:34, -0.68 for state:33 against its direct +1.05), 1390 (state:25 +0.39), 17102 (state:45 -0.66), 15490 -> 15904 (state:42 +0.54), 15362 (state:16 +0.38), 9954 (state:15 +0.37), 12690 (state:8 -0.20), 13068 (state:9 -0.14, plus direct -0.30), and 5533, which also reads day calls but is never active on train (pre-activation at most -0.30), so it contributes exactly 0. Evidence: sensitivities are on the matching state only (state:39 1.25/1.28, state:46 1.65, state:42 0.98); linear replacement AUC 0.8884 (unchanged); mean ablation 0.8856.

Closed form:

$$
\begin{aligned}
y_0 &= \operatorname{relu}\big(0.78 - 1.28x_{state:39_b}\big)\\
y_1 &= \operatorname{relu}\big(0.0571 + 1.02\operatorname{relu}\big(0.78\\
&\qquad - 1.28x_{state:39_b}\big)\big)\\
y_2 &= \operatorname{relu}\big(0.252 - 0.574x_{state:25}\big)\\
y_3 &= \operatorname{relu}\big(-0.38 + 1.01x_{state:15}\big)\\
y_4 &= \operatorname{relu}\big(1.36 + 1.65x_{state:46_b}\big)\\
y_5 &= \operatorname{relu}\big(-0.0337 + 1.0x_{state:8}\big)\\
y_6 &= \operatorname{relu}\big(-0.372 + 1.02x_{state:9_b}\big)\\
y_7 &= \operatorname{relu}\big(-0.581 + 0.999x_{state:16}\big)\\
y_8 &= \operatorname{relu}\big(0.361 + 0.984x_{state:42}\big)\\
y_9 &= \operatorname{relu}\big(-0.0356 + 0.993\operatorname{relu}\big(0.361\\
&\qquad + 0.984x_{state:42}\big)\big)\\
y_{10} &= \operatorname{relu}\big(-1.42 + 0.995x_{state:45}\big)\\
y_{11} &= \operatorname{relu}\big(0.383 + 0.163x_{state:33_b} - 1.5x_{state:26}\\
&\qquad - 0.732x_{state:34}\big)\\
y_{12} &= \operatorname{relu}\big(-1.54 + 0.655\operatorname{relu}\big(0.383\\
&\qquad + 0.163x_{state:33_b} - 1.5x_{state:26} - 0.732x_{state:34}\big)\\
&\qquad - 0.00543x_{totalDayCalls_b}\big)
\end{aligned}
$$

where y_0 is node `1295`; y_1 is node `7872`; y_2 is node `1390`; y_3 is node `9954`; y_4 is node `11361`; y_5 is node `12690`; y_6 is node `13068`; y_7 is node `15362`; y_8 is node `15490`; y_9 is node `15904`; y_{10} is node `17102`; y_{11} is node `5786`; y_{12} is node `5533`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0277, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Thirteen units, each reading only one-hot state indicators, so each is exactly an offset per state (least-squares R2 = 1.0): 1295 -> 7872 (state:39; weights -1.11, -1.25: a constant -2.32 logit for everyone except state:39, where it is switched off, +2.30, against state:39's direct -2.85; net -0.55 vs empirical log-odds offset -0.93, 82 customers), 11361 (state:46, +0.77 against direct -2.22), 578

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.5 A5: service_calls

Kind: Composition annotation (depth 1); parent A1 (section 1.6.1); children A6 (section 1.6.6), A7 (section 1.6.7).

- **Entry nodes:** `-71_ac` (numberCustomerServiceCalls_ac), `-46` (state:5), `-35` (state:4), `-23_b` (state:29_b), `-71_b` (numberCustomerServiceCalls_b).
- **Exit nodes:** `identity_svc`.
- **Internal nodes:** none.
- **Edges (weight):** `5268`→`identity_svc` (1.71), `9436`→`identity_svc` (1.88), `6713`→`identity_svc` (1.73).
- **With its descendants:** 11 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 10 candidate nodes.

Hypothesis. Composition of csc_early_hinge and csc_late_chain: their exits are summed by identity_svc = 1.88*9436 + 1.73*6713 + 1.71*5268, which enters the logit with weight 1; they read the same input (two copies of customer-service calls) plus state:29 (early hinge) and state:4/state:5 (late chain). Hypothesis: all of the model's service-call effect is a threshold: nothing for 0-2 calls, +0.30 logit at 3 calls, then about +2.3 logit per extra call (+2.06 at 4, +4.40 at 5, +6.67 at 6, +9.28 at 7); the early hinge carries most of it, the late chain adds an extra bend from 5 calls. For state:29 customers only the late chain remains. Evidence (train): churn by calls 0-7 is 0.115/0.110/0.110/0.106/0.437/0.571/0.679/0.545 against model 0.058/0.055/0.050/0.062/0.209/0.573/0.890/0.979: the threshold is right, but the model under-predicts up to 4 calls and is over-confident from 6 calls (42 customers). Sensitivity of identity_svc: state:29 1.04, calls (early copy) 0.48, state:4 0.13. Mean-ablating both children gives AUC 0.8016, linear replacement 0.8713; restoring service_calls alone onto the all-linear model gives 0.8657 (all-linear 0.8352).

Closed form (children as named functions):

$$\begin{aligned} y_0 = {} & 1.88\,\mathrm{csc_early_hinge_0}\big(x_{state:29_b}, \\ & x_{numberCustomerServiceCalls_b}\big) \\ & + 1.71\,\mathrm{csc_late_chain}\big(x_{numberCustomerServiceCalls_ac}, x_{state:5}, \\ & x_{state:4}\big) + 1.73\,\mathrm{csc_early_hinge_1}\big(x_{state:29_b}, \\ & x_{numberCustomerServiceCalls_b}\big) \end{aligned}$$

Expanded over the children:

$$\begin{aligned} y_0 = {} & 1.88\,\mathrm{relu}\big(-0.615 + 0.52x_{numberCustomerServiceCalls_b} \\ & - 1.12x_{state:29_b}\big) + 1.71\,\mathrm{relu}\big(-0.557 \\ & + 0.256\,\mathrm{relu}\big(0.476 + 0.74x_{numberCustomerServiceCalls_ac}\big) \\ & + 0.327x_{numberCustomerServiceCalls_ac} - 1.39\,\mathrm{relu}\big(0.219 \\ & + 0.414x_{state:4} - 1.86x_{state:5}\big)\big) + 1.73\,\mathrm{relu}\big(-0.0961 \\ & + 1.49\,\mathrm{relu}\big(-0.615 + 0.52x_{numberCustomerServiceCalls_b} \\ & - 1.12x_{state:29_b}\big)\big) \end{aligned}$$

where y_0 is node `identity_svc`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0157, and the exact function by at most 8.88×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).

- EMPIRICAL, `ablation_study`; data split: train. Composition of csc_early_hinge and csc_late_chain: their exits are summed by identity_svc = 1.88*9436 + 1.73*6713 + 1.71*5268, which enters the logit with weight 1; they read the same input (two copies of customer-service calls) plus state:29 (early hinge) and state:4/state:5 (late chain). Hypothesis: all of the model's service-call effect is a threshold: nothing for 0-2 calls, +0.30 logit at 3 ca

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1.6.6 A6: csc_late_chain

Kind: Leaf annotation (depth 2); parent A5 (section 1.6.5).

- **Entry nodes:** `-71_ac` (numberCustomerServiceCalls_ac), `-46` (state:5), `-35` (state:4).
- **Exit nodes:** `5268`.
- **Internal nodes:** `349`, `12374`.
- **Edges (weight):** `-35`→`349` (0.414), `-46`→`349` (-1.86), `349`→`5268` (-1.39), `12374`→`5268` (0.256), `-71_ac`→`5268` (0.327), `-71_ac`→`12374` (0.741).
- **Covers** (coverage, own and as part of the hierarchy): 5 candidate nodes.

Hypothesis. 12374 = relu(0.48 + 0.74 csc) (always active except at 0 calls, where it is just below its kink), 349 = relu(0.22 + 0.41 state:4 - 1.86 state:5), 5268 = relu(-0.56 + 0.33 csc + 0.26*12374 - 1.39*349); logit weight +1.71 (via identity_svc). 349 takes three values: 0.44 for most customers, 0 in state:5 (78 customers) and 4.55 in state:4 (41 customers). Hypothesis: a late hinge on customer-service calls that is off for 0-4 calls for most customers and adds about +0.65 logit per call from 5 calls (+0.55 at 5, +1.24 at 6, +1.88 at 7); in state:5 it starts one call earlier (on from 3 calls) and in state:4 it is always off. Evidence (train, n=4000): 5268 is active on 3.4% of customers (on for 3% at 4 calls, 99% at 5); sensitivity of 5268 is 0.074 to state:4, 0.017 to calls, 0.017 to state:5; line plot correlation 0.51. Mean ablation AUC 0.8889 (full 0.8884): it adds nothing to ranking on its own; linear replacement 0.8887 (R2 0.27); restored alone onto the all-linearised model 0.8401 (all-linear 0.8352).

Closed form:

$$
\begin{aligned}
y_0 = \operatorname{relu}\big(&-0.557 + 0.256\operatorname{relu}\big(0.476 \\
&+ 0.74x_{numberCustomerServiceCalls_ac}\big) + 0.327x_{numberCustomerServiceCalls_ac} \\
&- 1.39\operatorname{relu}\big(0.219 + 0.414x_{state:4} - 1.86x_{state:5}\big)\big)
\end{aligned}
$$

where y_0 is node `5268`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00148, and the exact function by at most 2.22×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 12374 = relu(0.48 + 0.74 csc) (always active except at 0 calls, where it is just below its kink), 349 = relu(0.22 + 0.41 state:4 - 1.86 state:5), 5268 = relu(-0.56 + 0.33 csc + 0.26*12374 - 1.39*349); logit weight +1.71 (via identity_svc). 349 takes three values: 0.44 for most customers, 0 in state:5 (78 customers) and 4.55 in state:4 (41 customers). Hypothesis: a late hinge on customer-service ca

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1.6.7 A7: csc_early_hinge

Kind: Leaf annotation (depth 2); parent A5 (section 1.6.5).

- **Entry nodes:** `-23_b` (state:29_b), `-71_b` (numberCustomerServiceCalls_b).
- **Exit nodes:** `9436`, `6713`.
- **Internal nodes:** none.
- **Edges (weight):** `9436`→`6713` (1.49), `-71_b`→`9436` (0.52), `-23_b`→`9436` (-1.12).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. 9436 = relu(-0.62 + 0.52 csc - 1.12 state:29), 6713 = relu(-0.10 + 1.49*9436); logit weights +1.88 and +1.73 (via identity_svc). Hypothesis: the main hinge on customer-service calls: off for 0-2 calls, just on at 3 calls (+0.30 logit), then about +1.8 logit per extra call (+2.04 at 4, +3.85 at 5, +5.43 at 6, +7.40 at 7); never on for the 72 state:29 customers. Active on 20.7% of customers (all with 3+ calls, outside state:29); empirical churn jumps from 0.11 at 3 calls to 0.44 at 4. Evidence: sensitivity 0.35 (6713) / 0.23 (9436) to state:29 and 0.16 / 0.11 to calls; the line_plot sweeps state:29 (first entry), so it is flat at 0 (tool quirk: line plots vary only the first entry). Mean ablation AUC 0.8281; linear replacement 0.8736 (R2 0.57/0.50); restored alone onto all-linear 0.8654.

Closed form:

$$
\begin{aligned}
y_0 &= \operatorname{relu}\big(-0.615 + 0.52 x_{numberCustomerServiceCalls_b} \\
&\qquad - 1.12 x_{state:29_b}\big) \\
y_1 &= \operatorname{relu}\big(-0.0961 + 1.49 \operatorname{relu}\big(-0.615 \\
&\qquad + 0.52 x_{numberCustomerServiceCalls_b} - 1.12 x_{state:29_b}\big)\big)
\end{aligned}
$$

where y_0 is node `9436`; y_1 is node `6713`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00608, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 9436 = relu(-0.62 + 0.52 csc - 1.12 state:29), 6713 = relu(-0.10 + 1.49*9436); logit weights +1.88 and +1.73 (via identity_svc). Hypothesis: the main hinge on customer-service calls: off for 0-2 calls, just on at 3 calls (+0.30 logit), then about +1.8 logit per extra call (+2.04 at 4, +3.85 at 5, +5.43 at 6, +7.40 at 7); never on for the 72 state:29 customers. Active on 20.7% of customers (all wit

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1.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft